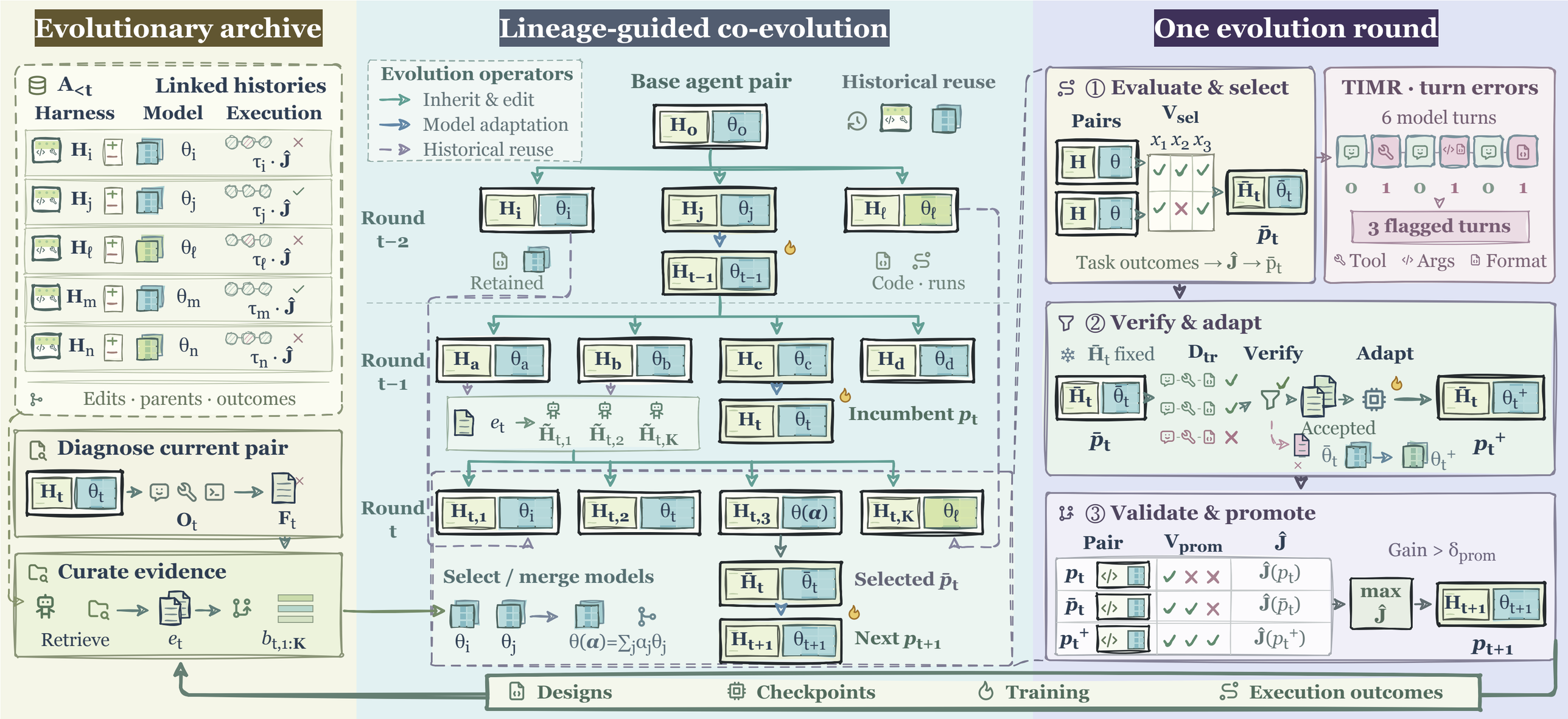

Evolutionary archive
$A_{<t}$
Linked histories
Harness
Model
Execution
H_i θ_i $\tau_i \cdot \hat{J}$
H_j θ_j $\tau_j \cdot \hat{J}$
H_ℓ θ_ℓ $\tau_\ell \cdot \hat{J}$
H_m θ_m $\tau_m \cdot \hat{J}$
H_n θ_n $\tau_n \cdot \hat{J}$
Edits · parents · outcomes
Diagnose current pair
H_t θ_t
O_t
F_t
Curate evidence
Retrieve
e_t
$b_{t,1:K}$
Lineage-guided co-evolution
Evolution operators
Inherit & edit
Model adaptation
Historical reuse
Base agent pair
H_o θ_o
Historical reuse
Round t−2
H_i θ_i
H_j θ_j
H_ℓ θ_ℓ
Retained
H_{t-1} θ_{t-1}
Code · runs
Round t−1
H_a θ_a
H_b θ_b
H_c θ_c
H_d θ_d
e_t → $\tilde{H}_{t,1}$ $\tilde{H}_{t,2}$ $\tilde{H}_{t,K}$
H_t θ_t
Incumbent p_t
Round t
$H_{t,1}$ θ_i
$H_{t,2}$ θ_t
$H_{t,3}$ $\theta(\alpha)$
$H_{t,K}$ θ_ℓ
Select / merge models
θ_i θ_j → $\theta(\alpha)=\sum_j \alpha_j \theta_j$
$\bar{H}_t$ $\bar{\theta}_t$
Selected $\bar{p}_t$
H_{t+1} θ_{t+1}
Next p_{t+1}
One evolution round
① Evaluate & select
Pairs
V_{sel}
x_1 x_2 x_3
H θ
H θ
$\bar{H}_t$ $\bar{\theta}_t$
$\bar{p}_t$
Task outcomes → $\hat{J}$ → $\bar{p}_t$
TIMR · turn errors
6 model turns
0 1 0 1 0 1
3 flagged turns
Tool
Args
Format
② Verify & adapt
$\bar{H}_t$ fixed
D_{tr}
Verify
Adapt
$\bar{H}_t$ $\bar{\theta}_t$
$\bar{p}_t$
Accepted
$\bar{\theta}_t$ → $\bar{\theta}_t^+$
$\bar{H}_t$ θ_t^+
p_t^+
③ Validate & promote
Pair
V_{prom}
$\hat{J}$
p_t ✓ ✗ ✗ $\hat{J}(p_t)$
$\bar{p}_t$ ✓ ✓ ✗ $\hat{J}(\bar{p}_t)$
p_t^+ ✓ ✓ ✓ $\hat{J}(p_t^+)$
Gain > δ_{prom}
max $\hat{J}$
H_{t+1} θ_{t+1}
p_{t+1}
Designs
Checkpoints
Training
Execution outcomes